

Course Syllabus

Operating Systems

ECOM 4401

1. Basic Data

Course Code and Title	ECOM 4401 - Operating Systems
Instructor	Belal K. Elfarra
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Primary Reference	A. Silberschatz, P. B. Galvin, and G. Gagne, Operating System Concepts, 10th ed., 2018.
Course Platform	Moodle Learning Management System (LMS)

2. Course Description

This course introduces the fundamental concepts, structures, and mechanisms of modern operating systems. It examines how an operating system manages processes, threads, processors, synchronization, deadlocks, main memory, virtual memory, storage, and file systems. The course combines conceptual understanding with practical exercises that help students observe and implement selected operating-system mechanisms using Linux and C.

3. Course Objectives

- Explain the role, services, and internal structure of an operating system.
- Describe processes and threads and explain how they are created, executed, and managed.
- Analyze and compare common CPU-scheduling algorithms.
- Explain process synchronization, race conditions, critical sections, mutexes, and semaphores.
- Describe deadlock conditions and compare methods for prevention, avoidance, detection, and recovery.
- Explain main-memory management and virtual-memory techniques.
- Analyze common page-replacement algorithms and their performance.
- Explain the basic organization of mass-storage and file systems.
- Apply selected operating-system concepts through practical Linux/C programming and simulation exercises.

4. Learning Resources and Course Management

The course will be managed through the Moodle Learning Management System (LMS). Lecture materials, recorded lectures, announcements, assignments, quizzes, project instructions, and examination information will be made available through the course page. **Students are expected to check Moodle regularly.**

Explained material will be available through the course homepage. Material not covered directly in lectures may be assigned for self-reading when it supports or extends the topics discussed in class.

Students should have access to a PDF copy of the course reference and the required lecture materials on their laptops.

5. Course Delivery and Communication

Course lectures will be delivered through recorded videos uploaded to Moodle. Students are required to watch the lectures and study the assigned material. Questions and discussions will be addressed during the instructor's announced online office hours.

All email correspondence should be written in clear, formal English or formal Arabic and should clearly state the purpose of the message.

6. Grade Distribution

Component	Marks (%)	Notes
Quizzes	10	
Assignments + Projects	20	
Midterm Exam	20	
Final Exam	50	Face-to-face on campus according to the university examination schedule.

7. Course Schedule

The sequence below follows the core topics identified in the original course plan. Exact dates may be adjusted according to the university academic calendar.

Week	Topic
Week 1	Introduction to Operating Systems
Week 2	Operating-System Structures
Week 3	Processes
Week 4	Threads
Week 5	CPU Scheduling
Week 6	Process Synchronization
Week 7	Deadlocks
Week 8	Main Memory
Week 9	Virtual Memory; selected topics from Mass-Storage Structure
Week 10	Virtual Memory and Page-Replacement Applications / Practical Exercises
Week 11	Mass-Storage and File-System Concepts / Practical Exercises
Week 12	File-System Interface and selected topics from File-System Implementation
Week 13	Project Work, Integration, and Selected Operating-System Topics
Week 14	Course Review and Project Completion

8. Practical Component / Project

The practical component is recommended to use Linux and the C programming language. It is intended to connect the theoretical material with observable operating-system behavior. Depending on available time, students may complete exercises or project tasks involving:

- Process creation and management using `fork()`, `exec()`, and `wait()`.
- Multithreaded programming using `pthread`.
- Synchronization using mutexes and semaphores.
- Simulation and comparison of CPU-scheduling algorithms.
- Simulation and comparison of page-replacement algorithms.
- Simulation or analysis of deadlock scenarios.

9. Supplementary / Self-Reading Topics

The following topics may be used for supplementary reading or brief enrichment activities rather than as core lecture material, depending on the available semester time:

- I/O Systems
- Advanced File-System Internals
- Security
- Protection
- Distributed Systems
- Virtualization
- Real-Time Operating Systems

10. Final Examination

The final examination will be closed book and will cover all explained course material. Concepts, figures, tables, algorithms, examples, and operating-system mechanisms discussed in the explained material may be used in the formulation of examination questions. The examination date, time, and room will be announced according to the university examination schedule.

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